

Predicting Mental Health Trends

Launch Report

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January 19th, 2025

1 Introduction

In recent years, mental health has gained significant global attention. Platforms like Reddit have become essential forums where individuals openly share their experiences, discuss challenges, and seek advice on mental health issues. These platforms house vast amounts of unstructured data, offering a unique opportunity to uncover insights into mental health trends, triggers, and coping mechanisms. By analyzing these discussions, we can better understand public sentiment and identify emerging patterns in mental health.

2 Deliverables

- **Data Acquisition:** Collection and preprocessing of Reddit data, including posts, comments, and engagement metrics from mental health-related subreddits.
- **Data Preprocessing:** Cleaning data by removing duplicates and noisy data. Text preprocessing includes tokenization and lemmatization.
- **Exploratory Data Analysis (EDA):** Detailed analysis of trends, examination of emotional tones in mental health discussions – visualizing them through graphs.
- **Final Report:** Documenting the data acquisition process and analysis which could contribute to increased awareness.

3 Highlighted Features

3.1 Gather Data Using Reddit API

- Use the Reddit API (via the PRAW library in Python, Google Colab) to collect data from mental health-focused subreddits, such as:

- r/anxiety
- r/depression
- r/adhd
- Extract post content, comments, timestamps, and other metadata for analysis.

3.2 Clean and Filter Data

- **Remove Irrelevant Data:** Eliminate posts or comments that do not contain meaningful text (e.g., URLs, advertisements, or automated messages).
- **Handle Missing Data:** Remove entries with missing or incomplete text fields.
- **Standardize Text:**
 - Convert all text to lowercase.
 - Remove special characters, numbers, and punctuation.
 - Expand contractions (e.g., "can't" to "cannot").
 - **Remove Stop Words:** Filter out common stop words (e.g., "and", "the", "is") using an NLP library like NLTK or SpaCy.
 - **Lemmatization/Stemming:** Reduce words to their root forms to standardize variations (e.g., *struggled* → *struggle*).

3.3 Create an Algorithm for Severity Scoring

- We will develop an algorithm to define severity levels for mental health discussions. This algorithm will be informed by keywords commonly associated with mental health, with severity levels determined based on insights from research papers and reputable sources.
- The algorithm will dynamically assign weights to these keywords, classifying them into high, medium, and low severity categories, ensuring the approach is evidence-based and reflective of current research.
 - **High Severity**
 - **Medium Severity**
 - **Low Severity**

3.4 Analyze Sentiment Over Time

- Track sentiment fluctuations across timeframes to identify trends in community well-being.
- Highlight periods of increased negative sentiment or high severity scores that might correlate with external factors (e.g., world events, holidays).
- Create dashboards or visual reports showing trends, sentiment shifts, and key discussion topics.
- Use machine learning to predict potential mental health trends based on past data.

4 Hybrid Learning Approach

To enhance the accuracy and flexibility of the analysis, we will adopt a hybrid learning approach that combines supervised and unsupervised methods.

4.1 Supervised Learning Topics

- **Severity Classification:** Train models to classify mental health discussions into predefined severity levels (e.g., high, medium, low).
- **Topic-Specific Classification:** Predict the primary mental health issue discussed in a post (e.g., anxiety, depression, ADHD).
- We will use:
 - **Random Forest:** Robust, non-linear classification with feature importance for interpreting severity indicators.
 - **XGBoost:** High accuracy in handling imbalanced datasets and boosting performance.
- For hyperparameter tuning, we will employ:
 - **Grid Search:** Systematic exploration of parameter space.
 - **Random Search:** Efficient sampling within the parameter space.

4.2 Unsupervised Learning Topics

- We will use techniques like **Latent Dirichlet Allocation (LDA)** to uncover latent themes in mental health discussions.
- Clustering methods will be applied to group similar posts based on their textual content or sentiment patterns.

5 Target Variable

We will define two primary target variables:

- The type of mental health disorder a post indicates.
- The severity level of the mental health condition based on post content.

6 Sponsor or Proxy User

None for now.

7 Issues

None for now.

8 Potential Issues in Data Acquisition for Predicting Mental Health Trends

8.1 Access to Data

- Reddit has strict guidelines; data scraping is subject to rate limits and may require multiple authentication restrictions.
- Some subreddits might have rules that restrict API access, requiring approval from moderators to access the data.

8.2 Data Quality

- Reddit posts often contain unstructured data, including typos, slang, or abbreviations, making it harder to clean and analyze.
- Some posts may be irrelevant to the mental health topic or contain spam, introducing noise that can affect analysis accuracy.

8.3 Limited Metadata

- Reddit data primarily includes basic details such as timestamps and engagement metrics but lacks demographic information (e.g., age, gender, location), which are essential for deeper insights into mental health trends.
- Self-reported data from users may be incomplete or inaccurate, making it difficult to derive precise conclusions.

9 Experts and Stakeholder Requirements

9.1 Professors and Faculty

- Professors from public health departments can help interpret mental health trends and provide relevant research resources.
- Data science professors can offer expertise in machine learning models and NLP techniques to analyze Reddit data effectively.

9.2 University Counselling Centre

- The university counselling center can provide anonymized information about common student mental health conditions.
- They can suggest important variables to focus on during the analysis.

9.3 Reddit Community Moderators

- Moderators can grant permission to access subreddit-specific data, refining the scope of the analysis.

10 Team Members and Their Specialties

10.1 Sanjoli Sogani (Team Lead)

- Will coordinate with all team members to ensure smooth collaboration and workflow.
- Will lead efforts in data pre-processing and cleaning to prepare the dataset for analysis and modeling.

10.2 Aman Ostwal

- Will be responsible for data acquisition by sourcing and preparing the required datasets.
- Will conduct Exploratory Data Analysis (EDA) to uncover insights and trends within the data.

10.3 Darshit Rai

- Will actively contribute to data acquisition by sourcing and preparing the required datasets.
- Will perform Exploratory Data Analysis (EDA) to identify patterns and inform further modeling processes.

10.4 Sai Pokuri

- Will play a pivotal role in data pre-processing and cleaning to ensure the dataset's readiness for analysis.
- Will lead Exploratory Data Analysis (EDA) to extract meaningful insights and trends.
- Will take charge of the modeling phase, designing and implementing predictive and analytical models.

11 Team Communication

11.1 Weekly Coordination on Teams

- At the beginning of each week, we communicate on Microsoft Teams to decide the meeting schedule based on everyone's availability.

11.2 Scheduled Meetings on Microsoft Teams

- On the scheduled day, we meet on Microsoft Teams to discuss tasks and deliverables.
- These meetings are brief but focused, ensuring that everyone understands their responsibilities and deadlines.

11.3 Mid-Week Follow-Up Meetings

- After the initial discussion, we plan another meeting later in the week on Teams to review progress and address any challenges.
- This ensures that tasks stay on track and any issues are resolved promptly.

12 File Sharing and Task Division on Teams

- All project-related documents and files are shared on Microsoft Teams.
- Tasks are divided among team members during the meeting, and everyone works on their assigned responsibilities.

13 Peer Review and Quality Check

- Once tasks are completed, team members review each other's work to ensure quality and consistency.
- This collaborative effort helps maintain ambitious standards and ensures that everyone is aligned with the project goals.

14 Team Issues

14.1 Data Acquisition Challenges

- Difficulty in collecting relevant Reddit data due to API limitations or restrictions.

14.2 Approaches

- Allocate time for troubleshooting API access issues and ensure familiarity with tools like PRAW or Pushshift API.

15 Appendix

Week	Plan	Result
Jan 06, 2025 – Jan 12, 2025	Project Selection	
Jan 13, 2025 – Jan 19, 2025	Finding Data Sources/data and setting up access to Reddit API for data collection	API setup complete and sample data fetched

Week	Plan	Result
Jan 20, 2025 – Jan 26, 2025	• Data collection from selected subreddits (e.g., anxiety, depression). • Store data systematically for easy access and processing.	• Dataset Collection. • Save in .csv format.
Jan 27, 2025 – Feb 02, 2025	• Perform data cleaning (e.g., removal of duplicates, null values, or irrelevant data). • Create a preprocessed dataset. • Set up a meeting with Counselling Centre Advisor to discuss which variables to focus on.	• Clean dataset ready for analysis.
Feb 03, 2025 – Feb 09, 2025	• Begin text preprocessing (e.g., tokenization, removal of stop-words, stemming, and lemmatization).	• Preprocessed dataset.
Feb 10, 2025 – Feb 16, 2025	• Work on presentation and data acquisition and pre-processing report.	• Presentation and report completed.
Feb 17, 2025 – Feb 23, 2025	• Conduct EDA on comments counts, votes. • Visualization of basic trends.	• Perform initial EDA.
Feb 24, 2025 – Mar 01, 2025	• Perform EDA on textual data (e.g., commonly used words, phrases, and sentiments). • Work on word frequencies, patterns, sentiment polarity.	• Prepare Sentiment Trend Visualization.

Week	Plan	Result
Mar 02, 2025 – Mar 08, 2025	• Task data severity level of each mental health problem. • Create Exploratory Data Analytics report.	• Complete EDA. • EDA report.
Mar 09, 2025 – Mar 15, 2025	• Refine the project based on feedback. • Create Status Presentation.	• Status Presentation.

16 Table of Contributions

The table below identifies contributors to various sections of this document.

	Section	Writing	Editing
1	Project	Everyone	Everyone
2	Team	Everyone	Everyone
3	Plan	Everyone	Everyone

Grading

The grade is given based on quality, clarity, presentation, completeness, and writing of each section in the report. This is the grade of the group. Individual grades will be assigned at the end of the term when peer reviews are collected.